

Day 04 of 60

YOU DON'T NEED TO MASTER PYTHON.

You need to master THIS SLICE.

Forget building web apps and games. In Data Science & ML, we only care about the **Data Stack**.

 4 Core Libraries

 1 Snippet Each

 Lean Stack

NumPy

Pandas

Matplotlib

Scikit-Learn



Vinod Bavage
Machine Learning · AI

 vinodbavage31

Day 04 / 60

Nu

NumPy (Numerical Python)

The absolute foundation of all ML math. Python lists are slow. NumPy arrays are written in C and are blazingly fast.

- Fast vector and matrix operations
- Mathematical functions (sin, cos, exp)
- Random number generation



numpy_demo.py

```
import numpy as np

# 1. Create a 2D matrix (Features)
X = np.array([[1, 2], [3, 4]])

# 2. Initialize random weights
W = np.random.randn(2, 2)

# 3. Fast matrix multiplication (Dot Product)
output = np.dot(X, W)

print(output.shape) # (2, 2)
```





Pandas

Built on top of NumPy, Pandas is the Excel of Python. It turns messy CSVs and SQL tables into clean DataFrames.

- Handling missing values
- Filtering, Grouping, and Merging data
- Exploratory Data Analysis (EDA)

```
pandas_demo.py

import pandas as pd

# 1. Load data from CSV
df = pd.read_csv('sales_data.csv')

# 2. Clean and aggregate in one chain
clean_data = (
    df.dropna(subset=['price'])
      .groupby('category')['revenue']
      .mean()
)

print(clean_data.head())
```





Matplotlib

Numbers are hard to digest. Plots uncover trends. Matplotlib (and Seaborn) lets you visualize data distributions easily.

- Scatter plots to find correlations
- Histograms to see data spread
- Visualizing model performance

```
plot_demo.py

import matplotlib.pyplot as plt

# 1. Create a scatter plot
plt.scatter(df['age'], df['income'], c='cyan')

# 2. Add labels for readability
plt.title("Age vs Income Correlation")
plt.xlabel("Age (Years)")
plt.ylabel("Income ($)")

# 3. Render the plot
plt.show()
```





Scikit-learn

The Swiss Army Knife of Machine Learning. It provides consistent APIs for hundreds of algorithms out of the box.

- Regression, Classification, Clustering
- Data preprocessing & scaling
- Model evaluation metrics

```
model_demo.py

from sklearn.ensemble import
RandomForestClassifier

# 1. Initialize the model
model =
RandomForestClassifier(n_estimators=100)

# 2. Train the model on data
model.fit(X_train, y_train)

# 3. Predict on unseen data
predictions = model.predict(X_test)

# Machine Learning in 4 lines!
```



04

✓ Day 04 Complete

Tomorrow on Day 05:

Linear Regression Intuition

How the simplest ML model works under the hood — without complex math.

Follow for the full 60-day ML series

Vinod Bavage

Building in public · Machine Learning · 60 days

#Python

#MachineLearning

#DataScience

#VinodBavage

